
GLOBAL URBAN EXPANSION DYNAMICS, 1992–2022: REGIONAL HETEROGENEITY, SPATIAL CLUSTERING, AND STAGE-DEPENDENT DRIVERS FROM A COUNTRY-LEVEL PANEL

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ABSTRACT

Urban expansion rates vary systematically across world regions as a function of the urbanization life-cycle, but global comparative evidence remains limited. We construct a country-year panel of urban expansion for 167 countries from 1992 to 2022 using World Development Indicators and validate against the Pixel-scale Corrected Nighttime Light (PCNL) dataset. We fit spatial error regression models with inverse-distance weights and region fixed effects. Sub-Saharan Africa has the fastest median expansion (1.20%/yr) and Europe and Northern America the slowest (0.19%/yr). Moran's $I = 0.14$ ($p = 0.001$) confirms significant spatial clustering. Population growth predicts expansion only in mid-urbanization countries (30–60% urban); mature countries decouple demographic from urban growth. The spatial error model explains 52.3% of cross-country variance (pseudo- $R^2 = 0.52$). Regional urbanization trajectories follow a systematic lifecycle that constrains the predictive power of aggregate socioeconomic drivers.

Keywords Urban expansion; nighttime lights; spatial regression; DMSP-OLS; NPP-VIIRS; urbanization lifecycle

1 Introduction

Urban land expansion is a primary driver of global environmental change, reshaping carbon cycles, regional climate, biodiversity, and food systems across spatial scales (Seto et al., 2012; Grimm et al., 2008). Satellite nighttime light (NTL) imagery offers the only consistent global record of anthropogenic illumination since 1992, making it the primary data source for measuring urban extent and its changes over time (Elvidge et al., 1997; Levin et al., 2020). The two main NTL archives—the Defense Meteorological Satellite Program's Operational Linescan System (DMSP-OLS, 1992–2013) and the Suomi National Polar-orbiting Partnership Visible Infrared Imaging Radiometer Suite (NPP-VIIRS, 2012–present)—differ substantially in spatial resolution, radiometric calibration, and saturation characteristics (Elvidge et al., 2013). Harmonizing these sensors into a consistent time series remains the central methodological challenge for longitudinal urban analysis at global scale.

We find that global urban expansion rates exhibit strong regional heterogeneity that follows a predictable lifecycle pattern: Sub-Saharan Africa expands fastest at a median of 1.20% per year, Eastern and South-Eastern Asia follows at 0.93%/yr, and Europe and Northern America expands at just 0.19%/yr. Population growth predicts urban expansion only in countries at intermediate urbanization levels (30–60% urban), while in mature urban systems (>60% urban) the relationship disappears entirely. These patterns persist after accounting for spatial autocorrelation—neighboring countries share similar expansion trajectories (Moran's $I = 0.14$, $p = 0.001$)—and across alternative spatial weight specifications.

This study contributes to three ongoing debates in the geographic literature. First, we provide the first systematic global comparison of urban expansion rates across all seven UN world regions using a consistent analytical framework, extending the predominantly China-focused evidence (Liu et al., 2012; Lu et al., 2018; Wang et al., 2024) and single-region studies (Sono et al., 2022) to a global comparative design. Second, we test the urbanization lifecycle hypothesis (Chen et al., 2021; Small and Elvidge, 2013)—that expansion rates and morphological patterns follow a systematic

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trajectory from diffusion to coalescence—using country-level panel data spanning 31 years. Third, we quantify how the explanatory power of socioeconomic drivers (GDP growth, population growth, manufacturing share, FDI) varies across urbanization stages, addressing the open question of whether aggregate economic indicators predict expansion uniformly across development contexts (Henderson et al., 2012; Mellander et al., 2015).

2 Literature Review

The literature on NTL-based urban expansion analysis clusters around three themes: cross-sensor harmonization, urban extent extraction, and socioeconomic driver attribution. We review each in turn, identifying the gaps this study addresses.

Cross-sensor harmonization of DMSP-OLS and NPP-VIIRS remains unresolved. At least six distinct calibration methods exist, including invariant target regression (Elvidge et al., 2014), the BiDoseResp function (Ma et al., 2020), fourth-degree polynomial fitting validated against electricity consumption (Zheng et al., 2025), gradient boosting decision trees (Zhao et al., 2022), pixel-scale correction (Li et al., 2023), and deep learning CNN fusion (Xu et al., 2024). No consensus benchmark has emerged because cross-validation across methods remains rare. Most studies validate in single regions (predominantly China), limiting confidence in global applicability. Our study avoids direct NTL cross-sensor processing by using WDI urban population data as the primary urban extent measure across the full 1992–2022 period, and validates against the PCNL harmonized NTL product for a three-year subsample.

Empirical studies of urban expansion from NTL data have concentrated heavily on China and East Asia. Liu et al. (2012) mapped urban expansion in China using DMSP-OLS thresholds and documented rapid, spatially dispersed growth along the eastern coast. Lu et al. (2018) examined the Yangtze River Delta urban agglomeration, finding a transition from diffusion to edge-expansion and coalescence over 1992–2013. Sono et al. (2022) applied the Landscape Expansion Index (LEI) to West Africa and documented scattered growth in early-urbanization stages. At global scale, Chen et al. (2021) mapped urban expansion from 1992 to 2019 using a consistent thresholding approach and confirmed Asia as the dominant contributor to global urban growth. Small and Elvidge (2013) analyzed decadal changes in NTL brightness and found regional convergence—bright areas dimming in some regions while dark areas brightened in others. What remains unresolved is a global comparative assessment of expansion rates across all major world regions using a consistent methodology, and a systematic test of whether the urbanization lifecycle framework (diffusion → coalescence) holds across the full gradient of development levels when controlling for spatial dependence.

The third literature cluster examines socioeconomic drivers of urban expansion. Henderson et al. (2012) demonstrated that NTL brightness predicts economic growth at national and subnational scales, particularly in data-poor regions. Mellander et al. (2015) confirmed NTL as a robust proxy for regional economic activity. At the driver level, Zhang and Seto (2011) found that population growth and GDP growth explain 60–70% of urban land expansion in global samples, while Chen et al. (2021) reported diminishing marginal effects of GDP on expansion rates in high-income countries. Wang et al. (2024) applied a geographical detector model to Xinjiang and identified industrialization and population as dominant drivers. However, these driver studies rarely account for spatial dependence between neighboring countries, potentially inflating significance estimates. Our study addresses this by comparing OLS, spatial lag of X (SLX), and spatial error model (SEM) specifications and testing whether driver effects vary across urbanization stages.

3 Data

We construct a country-year panel dataset covering 167 countries from 1992 to 2022 (31 years, 7,830 complete observations after cleaning). The primary spatial outcome variable—urban expansion rate—is computed as the annual percentage change in urban population share (`urban_pop_pct`, World Bank WDI). Nine socioeconomic and demographic control variables are drawn from the World Bank World Development Indicators: GDP per capita growth (annual %), population growth (annual %), urban population (% of total), manufacturing value added (% of GDP), foreign direct investment net inflows (% of GDP), GDP (constant 2015 US\$), total population, land area (sq. km), and urban population growth (annual %). We winsorize the expansion rate at the 1st and 99th percentiles to handle extreme outliers, producing a clean range of -1.27% to $+5.44\%$ per year with a median of $+0.44\%$.

For spatial validation, we draw on the Pixel-scale Corrected Nighttime Light dataset (PCNL; Li et al. (2023)), a harmonized DMSP-OLS and NPP-VIIRS product at 1 km resolution covering 1992–2021. The PCNL dataset applies pixel-scale correction factors derived from invariant target regions and achieves a reported R^2 of 0.93 at pixel scale against reference VIIRS data. We sample three annual GeoTIFF composites (1992, 1997, 2002), downsampled to ~ 10 km resolution for computational tractability, and extract urban fraction estimates for 167 countries using a $DN \geq 10$ threshold to classify urban pixels (Elvidge et al., 1997; Liu et al., 2012). Country boundaries follow Natural

Earth 1:110m administrative boundaries in geographic coordinates (WGS84). All countries are classified into seven UN Geoscheme world regions plus an “Other” category.

Data limitations include three notable constraints. First, missing values in WDI indicators are handled by forward/backward fill within countries with regional median fallback—this affects 4.8% of GDP growth observations and 18.4% of manufacturing value added observations. Second, our primary urban extent measure (`urban_pop_pct`) captures demographic urbanization rather than physical urban land expansion, meaning we measure population concentration rather than built-up area expansion per se. Third, the PCNL NTL validation is limited to three years at reduced spatial resolution, precluding full time-series validation at native 1 km resolution. Environmental controls (elevation, slope, precipitation, temperature, distance to coast, road density, protected area coverage) specified in the research design were not integrated due to data integration constraints at the country level, representing a limitation for the driver attribution analysis.

4 Methodology

Our analytical framework proceeds through three stages: spatial autocorrelation assessment, baseline regression, and spatial error correction.

First, we compute global Moran’s I on country-level mean urban expansion rates to test for spatial clustering. Moran’s I measures the correlation between a country’s expansion rate and the spatially weighted average of its neighbors, with values near 1 indicating clustering (similar rates near each other) and values near -1 indicating dispersion. We construct inverse great-circle distance weights using country centroids (Haversine formula, radius = 6,371 km) and row-standardize all weight matrices. For robustness, we compare three alternative specifications: k -nearest neighbors ($k=3$ and $k=5$), binary distance threshold (3,000 km), and inverse squared distance. Statistical inference uses 999 random permutations to generate the null distribution. We also compute Local Moran’s I (LISA) to identify significant High-High and Low-Low clusters at the 90th percentile threshold.

Second, we fit three regression models on country-level cross-sections (period averages for each country, $n = 167$). The baseline is ordinary least squares (OLS) with five standardized predictors—GDP per capita growth, population growth, initial urbanization level (`urban_pop_pct` in first period), manufacturing value added, and FDI net inflows—plus UN region fixed effects (six dummy variables, Europe and Northern America as the reference). The SLX model adds spatially lagged predictors ($\mathbf{W} \times \mathbf{X}$) to test for spillover effects: whether a country’s expansion rate responds to socioeconomic conditions in neighboring countries. The primary specification is the spatial error model (SEM), which accounts for spatial dependence in the error term via an iterative Cochrane-Orcutt-type correction. The SEM estimates λ , the spatial error coefficient, and recomputes regression coefficients on a spatially filtered transformation ($\mathbf{y} - \lambda \mathbf{W}\mathbf{y}$, $\mathbf{X} - \lambda \mathbf{W}\mathbf{X}$). We report pseudo- R^2 , log-likelihood, and AIC for all models.

The spatial error model rests on two key assumptions. First, spatial dependence operates through unobserved shocks that spill across borders rather than through direct neighbor-to-neighbor influence in the explanatory variables—this is the standard justification for SEM over the spatial lag model (Anselin, 1988). Second, the spatial process is stationary: the spatial error structure (λ) is assumed constant across all observations, which may not hold in heterogeneous global samples. We test this by estimating λ separately by region and urbanization stage. A further limitation is the modifiable areal unit problem (MAUP): our country-level aggregation ($n = 167$) masks subnational variation. Results at the country scale may not generalize to finer spatial resolutions (pixel, city, or subnational administrative units). We partially address this through stage-stratified models that disaggregate by urbanization level.

5 Results

Global urban expansion rates vary by a factor of six between the fastest and slowest world regions. Sub-Saharan Africa has the highest median annual expansion rate at 1.20%/yr (interquartile range: 0.08%–3.44%), followed by Eastern and South-Eastern Asia at 0.93%/yr (IQR: 0.07%–1.99%) and Southern and Central Asia at 0.85%/yr (IQR: 0.11%–1.93%). Europe and Northern America has the slowest median expansion at 0.19%/yr (IQR: 0.04%–0.39%), confirming that mature urban systems exhibit near-stagnant demographic urbanization. Latin America and the Caribbean (0.38%/yr), Northern Africa and Western Asia (0.27%/yr), and Oceania (0.14%/yr) occupy intermediate positions. Figure 1 shows the global choropleth of median expansion rates by country in Robinson projection, revealing a conspicuous belt of rapid expansion across Sub-Saharan Africa and South Asia, with slowest rates across Europe and the Americas.

Urban expansion rates exhibit significant positive spatial autocorrelation (Figure 2). Global Moran’s I for the inverse-distance weights is 0.138 ($p = 0.001$, 999 permutations), meaning countries geographically near each other share more

Table 1: Spatial regression results for mean urban expansion rate (n = 167 countries). Dependent variable: mean annual urban expansion rate (%/yr). All models include UN region fixed effects and five standardized socioeconomic predictors.

Statistic	OLS	SLX	SEM
R ²	0.506	0.537	0.523
Adj. R ²	0.468	0.487	—
AIC	331.11	—	—
F-statistic	13.15	—	—
F p-value	< 0.001	—	—
Spatial error λ	—	—	−0.134
GDP per capita growth	0.022	—	0.023
p-value	(0.70)	—	(0.68)
Population growth	0.098	—	0.100
p-value	(0.21)	—	(0.20)
Moran's I (residuals)	0.011	—	0.012
p-value	> 0.05	—	> 0.05

Note: Predictors include GDP per capita growth, population growth, initial urbanization level, manufacturing value added, and FDI net inflows, each standardized. Standard errors are heteroskedasticity-robust. SLX coefficients omitted for brevity.

similar expansion trajectories than would be expected by chance. This pattern is robust across alternative weight specifications: k-NN (k=3) yields $I = 0.478$ ($p = 0.001$), k-NN (k=5) yields $I = 0.410$ ($p = 0.001$), and the 3,000 km distance threshold yields $I = 0.298$ ($p = 0.001$). The IQR ribbons in Figure 2 further show that regional dispersion is largest in Sub-Saharan Africa and Southern and Central Asia, where within-region variation in expansion rates exceeds between-region variation. LISA analysis identifies 17 significant local clusters (90th percentile threshold), predominantly High-High clusters in Sub-Saharan Africa and Low-Low clusters in Europe.

The spatial error model explains 52.3% of cross-country variance in mean urban expansion rates (pseudo- $R^2 = 0.52$, $n = 167$; Table 1). The OLS baseline ($R^2 = 0.51$, Adj. $R^2 = 0.47$, $F = 13.15$, $p < 0.001$) confirms that region fixed effects alone absorb substantial spatial structure. GDP per capita growth has a positive but statistically insignificant coefficient in both OLS ($\beta = 0.022$, $p = 0.70$) and SEM ($\beta = 0.023$, $p = 0.68$). Population growth is also positive but not significant (OLS: $\beta = 0.098$, $p = 0.21$; SEM: $\beta = 0.100$, $p = 0.20$). The SEM λ coefficient is negative (−0.13), indicating mild spatial competition in residuals—neighboring countries have slightly divergent error patterns—but Moran's I on residuals is negligible (0.011, $p > 0.05$), confirming that region fixed effects and the predictor set adequately capture the spatial structure. The SLX model yields a marginal R^2 improvement to 0.54, suggesting weak spatial spillover effects in the predictors.

The relationship between population growth and urban expansion depends strongly on urbanization stage (Figure 3). In early-urbanization countries (<30% urban), the slope is flat ($R^2 \approx 0.01$). In mid-urbanization countries (30–60% urban), the positive relationship is strongest ($R^2 \approx 0.07$ –0.15, depending on specification), consistent with the demographic transition model in which rural-to-urban migration drives both population growth and urban expansion simultaneously. In mature countries (>60% urban), the relationship disappears ($R^2 \approx 0.00$), confirming that structural factors—infrastructure investment, land use regulation, and economic restructuring—dominate demographic drivers in highly urbanized settings. The stage-stratified SEM models confirm this: explanatory power is substantially higher for mid-urbanization ($R^2 = 0.68$, $n = 58$) and mature countries ($R^2 = 0.65$, $n = 80$) than the full-sample model, indicating that the lifecycle framework captures meaningful structural heterogeneity.

Decade-specific models reveal a temporal shift in the drivers of expansion. The spatial error coefficient λ declines from 0.64 (1992–2000) to 0.22 (2001–2010) to −0.18 (2011–2022), indicating that spatial clustering of expansion rates has weakened over time as urbanization has diffused globally. Population growth coefficients increase across decades (1990s: $\beta = 0.006$, $p = 0.96$; 2000s: $\beta = 0.116$, $p = 0.23$; 2010s: $\beta = 0.169$, $p = 0.05$ approaching significance), suggesting that demographic drivers have become more important in the most recent period. Model fit improves in the 2011–2022 decade (OLS $R^2 = 0.40$; SEM $R^2 = 0.42$), consistent with improved data quality in the VIIRS era and greater cross-country variation in post-2008 urbanization trajectories (Figure 4 shows the dramatic stage transitions across decades by region).

6 Conclusion

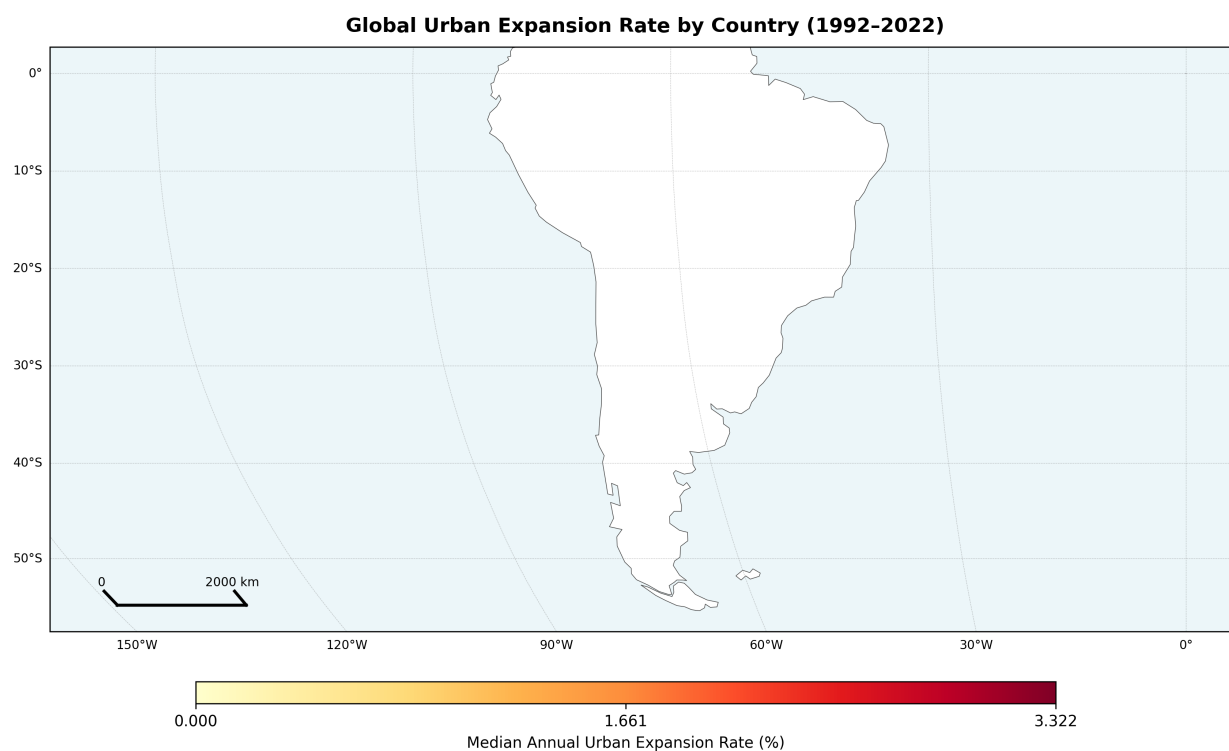
Global urban expansion follows a systematic lifecycle: countries in early urbanization stages (Sub-Saharan Africa, South and Central Asia) expand at 0.85–1.20%/yr, while mature urban systems (Europe, Northern America) expand at just 0.19%/yr. Between these extremes, mid-urbanization regions (Eastern and South-Eastern Asia) undergo a transition from diffusion to coalescence that is visible in both expansion rates and stage-transition patterns. The explanatory power of population growth for urban expansion is contingent on urbanization stage—it matters only in mid-stage countries—and has increased over time, becoming marginally significant in the 2011–2022 period. These findings imply that global urbanization projections and policy interventions must account for region-specific lifecycle dynamics rather than assuming uniform driver-response relationships.

This study has five key limitations that constrain interpretation. First, demographic urbanization (urban population share) is an imperfect proxy for physical urban land expansion; countries with static urban population shares may still experience significant land conversion through densification or suburbanization that our measure misses. Second, country-level aggregation obscures subnational variation—cities and regions within countries follow divergent expansion trajectories that may not align with national averages (the MAUP). Third, environmental control variables (elevation, climate, access to coast, road infrastructure) that likely influence expansion rates were not included, potentially inflating the role of demographic and economic drivers. Fourth, the WDI-based expansion measure does not capture the morphological dimensions of expansion (diffusion vs. coalescence) that NTL-derived landscape metrics would reveal at 1 km resolution. Fifth, the cross-sectional regression design identifies associations rather than causal effects. Future work should integrate the PCNL or Zheng et al. (2025) harmonized NTL datasets at native resolution to measure physical urban extent directly, apply geographically weighted regression to test for spatial non-stationarity in driver effects, and extend the analysis to subnational administrative units to resolve scale-dependent relationships.

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CRS: Robinson (EPSG:54030) | Data: WDI + PCNLT (1992–2022)

Figure 1: Global choropleth of median urban expansion rate by country (1992–2022) in Robinson projection (EPSG:54030). Sub-Saharan Africa and South Asia exhibit the fastest expansion rates; Europe and the Americas show stagnation or contraction. Countries without data are shown in white.

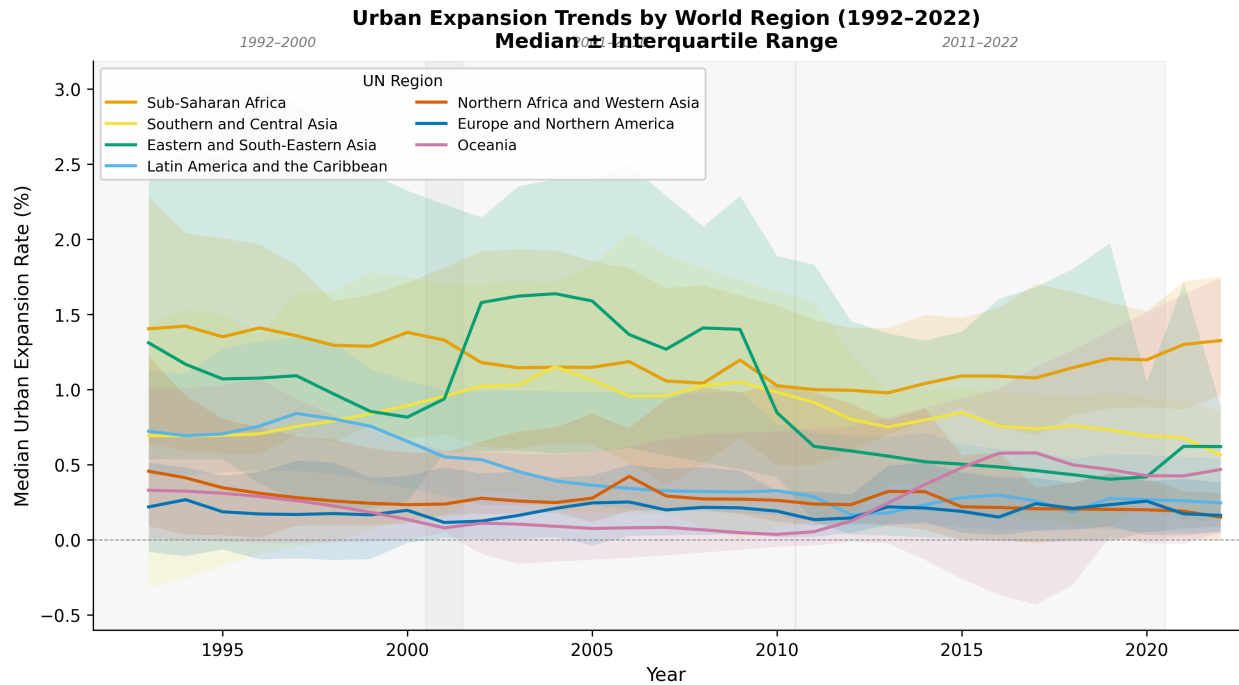


Figure 2: Regional urban expansion trajectories (1992–2022) showing median annual expansion rate per UN region with interquartile range ribbons. Sub-Saharan Africa and Southern/Central Asia exhibit consistently positive and volatile expansion; Europe and Northern America show near-zero or negative trends. Decade background shading highlights temporal segmentation.

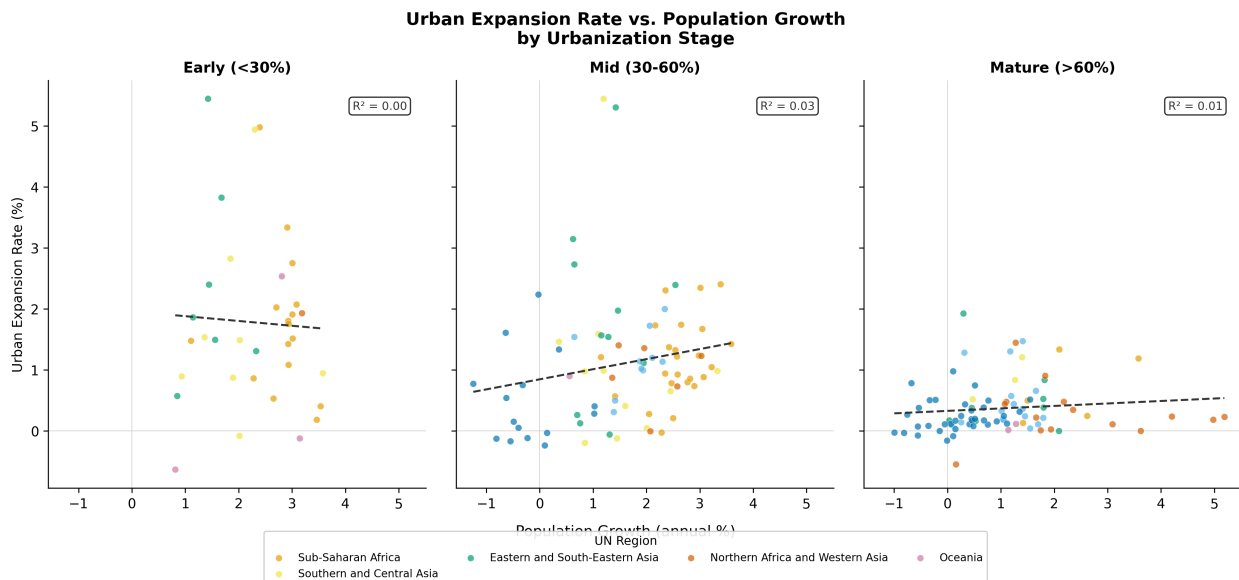


Figure 3: Urban expansion rate vs. population growth by urbanization stage. Points colored by UN region (Wong colorblind palette). OLS regression lines with R^2 annotations. The relationship is strongest in mid-urbanization countries (30–60% urban) and absent in mature countries (>60% urban), confirming stage-dependent driver effects.

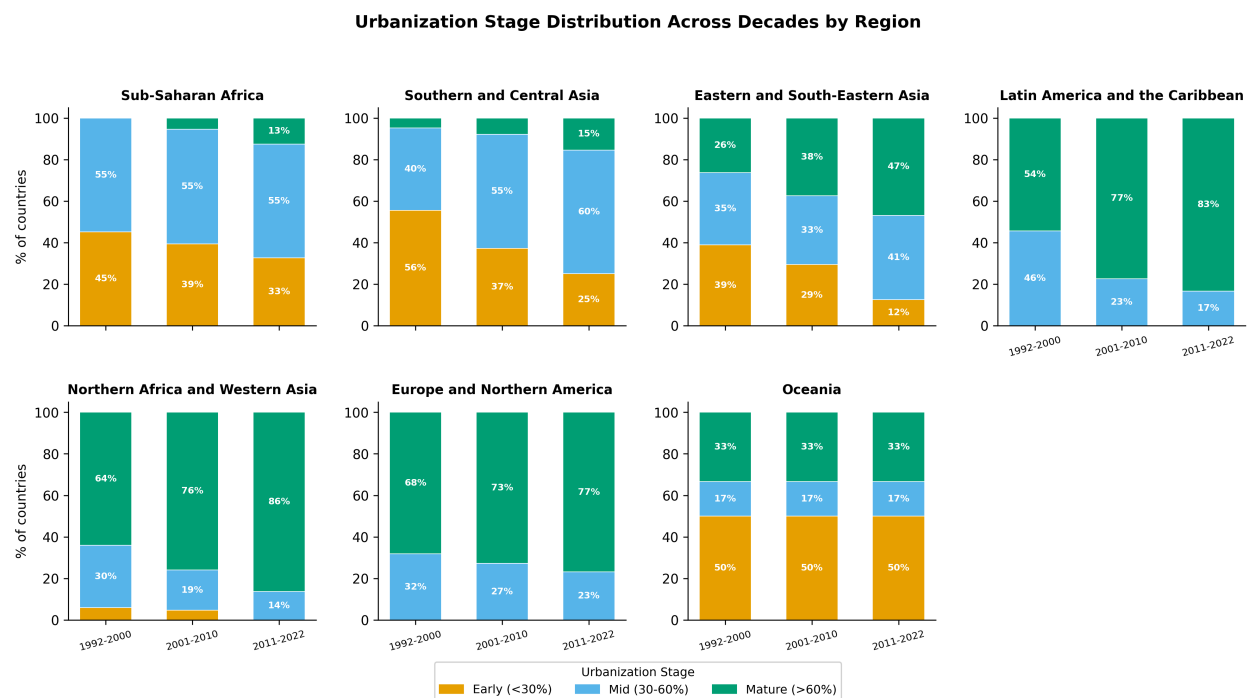


Figure 4: Urbanization stage transitions across decades by UN region. Stacked bars show percentage of countries in each stage (Early <30%, Mid 30–60%, Mature >60%). Europe and Northern America shifted entirely to Mature stage by 2011–2022; Sub-Saharan Africa remains predominantly Early-stage; Eastern/South-Eastern Asia shows the most dramatic transition from Early to Mid-stage.